Multi-Agent Reinforcement Learning for Condition-Based Textile Sorting with a Tendon-Driven and Collaborative Robot System

Master Thesis im Studiengang Computational Engineering M. Sc.

Friedrich-Alexander-Universität Erlangen-Nürnberg
Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
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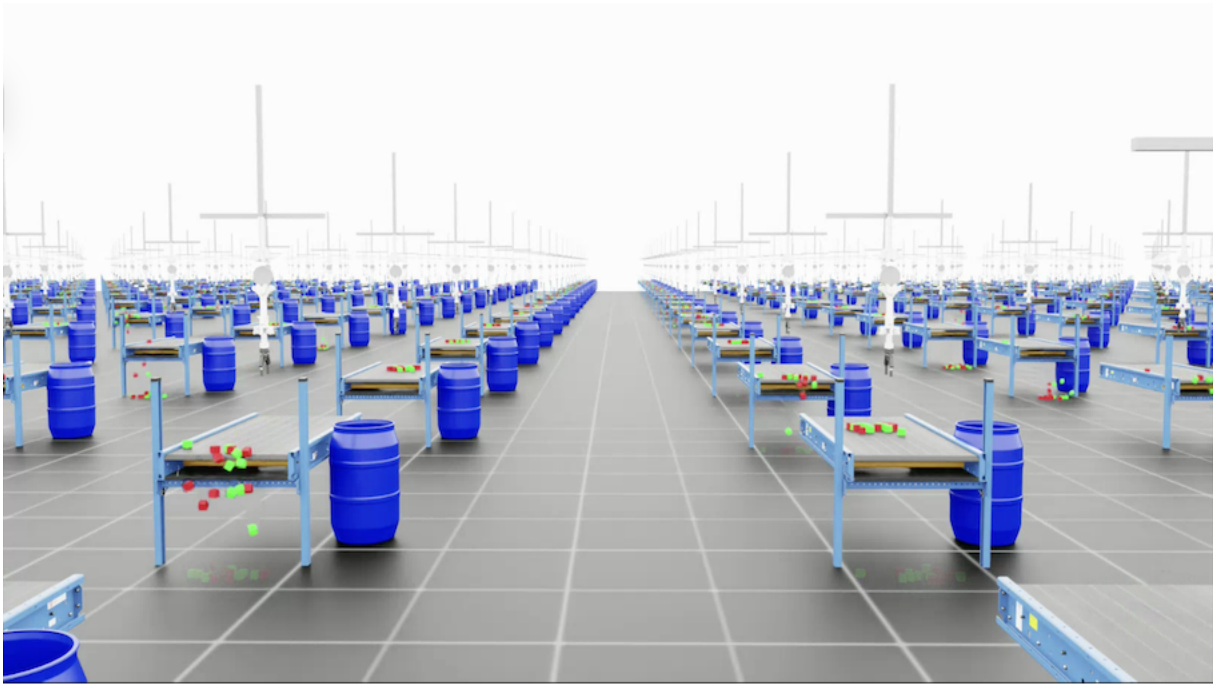

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Abgabetermin:	TBD
Bearbeitungszeit:	x Monate

Erklärung

Ich versichere, dass ich die vorliegende Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat und von dieser als Teil einer Prüfungsleistung angenommen wurde. Alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Erlangen, den 12. März 2026

Georg Meyer

Contents

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List of Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
CPU	Central Processing Unit
DoF	Degree of Freedom
DR	Domain Randomization
dVRK	da Vinci Research Kit
FEM	Finite Element Method
FK	Forward Kinematics
GAE	Generalised Advantage Estimation
GPU	Graphics Processing Unit
IK	Inverse Kinematics
IL	Imitation Learning
MDP	Markov Decision Process
MJCF	MuJoCo XML Format
MPC	Model Predictive Control
MPM	Material Point Method
PBD	Position-Based Dynamics
PD	Proportional-Derivative
PGS	Projected Gauss–Seidel
PID	Proportional-Integral-Derivative
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RGB-D	Red-Green-Blue plus Depth
RL	Reinforcement Learning
ROS	Robot Operating System
SDF	Signed Distance Field
TGS	Temporal Gauss–Seidel
URDF	Unified Robot Description Format
USD	Universal Scene Description
VBD	Vertex Block Descent
XML	Extensible Markup Language
XPBD	Extended Position-Based Dynamics

List of Symbols

$\mathcal{A}$	Action space
$\hat{A}_t$	Advantage estimator
ϵ	Clipping parameter (PPO)
γ	Discount factor
h	Simulation time step
$J(\theta)$	Expected discounted return
k_j	Stiffness parameter (PBD)
$\mathbf{M}$	Mass matrix
π_θ	Policy (parameterized by θ)
$r(s_t, a_t)$	Reward function
$\mathcal{S}$	State space
θ	Policy parameters

0 Guideline

0.1 Format and Style

0.1.1 Page Layout and Typography

- Paper format: white DIN A4 paper
- Font: Arial, 12pt (in this LaTeX template, the sans-serif option achieves this)
- Line spacing: 1.2 (the template uses 1.25)
- Text alignment: justified (Blocksatz) with automatic hyphenation enabled
- Font emphasis: use only in exceptional cases; headings follow the template style and must not be underlined
- Bullet points: only square bullet points (FAPS green), never round ones; use the template's built-in list style
- FAPS color scheme: FAPS-Grün (RGB 151, 193, 57) and white

0.1.2 Page Numbering and Binding

- Page numbering starts with the introduction (Arabic numerals)
- Directories (table of contents, list of figures, etc.) receive separate Roman numeral numbering; the cover page is not counted
- Print single-sided unless otherwise agreed with the supervisor
- Two bound copies required: adhesive binding with clear cover foil and green (FAPS green) back sheet
- Each copy must include a CD/DVD containing:
  - Digital version of the thesis (PDF)
  - All digital sources (papers, scanned documents, websites)
  - All images, tables, measurement data, models, and other data

0.1.3 Section and Chapter Formatting

- Chapters are numbered numerically with at most three levels (e.g., 1.1.1); no trailing dot after the last digit

- If a subsection X.1 exists, there must also be at least an X.2 (applies to all levels)
- Main chapters should have at least 3 subchapters; aim for a balanced structure (e.g., 5 chapters at level 1, each with 4 at level 2, each with 3 at level 3)
- Headings in the table of contents must match the headings in the text verbatim
- Every heading must be followed by actual content (including introductory text at the start of main chapters)

0.1.4 Paragraphs, Spacing, and Whitespace

- Each paragraph must consist of at least two sentences
- No blank line between paragraphs within a chapter; one blank line before the next heading at the end of a chapter
- No double spaces anywhere in the document; use search-and-replace to eliminate them before submission
- Use non-breaking spaces (in LaTeX: ˜) between:
  - Titles and names: “Prof. Dr. Beispielhausen”
  - Date expressions: “19. September 2007”
  - Abbreviations: “z. B.”, “u. a.”, “d. h.”
  - Numbers and units: “44 mm”
  - Numbers and words: “1.350 Personen”, “Nr. 5”
  - Citation references
- Check automatic hyphenation carefully for errors (e.g., incorrect compound-word breaks)

0.1.5 Figures

- Every figure receives a caption below it (Unterschrift), formulated so the figure is understandable without the surrounding text
- All figures are numbered consecutively with Arabic numerals and listed in the list of figures
- Every figure must be referenced in the body text
- Title image: no caption, no numbering, not included in the list of figures
- Appendix figures: numbered within each appendix separately (e.g., Appendix A – Figure 1); not included in the list of figures
- Ensure sufficient image quality; recreate figures if originals are blurry, low-resolution, or scanned
- Use the same font in figures as in body text (Arial); font size 10pt is acceptable but must be consistent across all figures

- All figure components must be fully labeled: photo components, axes, parameters, legends with complete process parameters
- Fill areas with color only if the color conveys meaningful information; otherwise keep figures clean and minimal
- For recreated/modified figures from sources, cite with “in Anlehnung an [xy]” (based on [xy])
- Use a consistent color scheme across all figures; preferably FAPS colors unless industry cooperation requires otherwise
- Use a self-created graphic on the first page of the thesis

0.1.6 Tables

- Every table receives a caption above it (Überschrift) with a meaningful, descriptive title
- Tables are numbered consecutively with Arabic numerals and listed in the table directory
- Every table must be referenced in the body text
- Use the same font as in body text; maintain consistent font size across all tables
- Preferred alignment inside tables: left-aligned or centered (not justified)
- If a table spans a page break, do not break within a row; repeat the table header on the new page
- Extensive tables belong in the appendix

0.1.7 Diagrams

- Follow DIN 461 for axis labeling in diagrams
- Label axes with an arrow in the direction of increase (except for trial numbers etc.)
- Replace the axis label at the second-to-last tick mark with the unit
- Display all relevant information within the diagram
- Place a frame around diagrams
- Use Arial font inside diagrams with a legible font size
- Diagrams are captioned and numbered as figures (no separate numbering type)

0.1.8 Formulas

- Formula numbers are enclosed in round parentheses

- Numbering is chapter-based in the form X.Y (X = chapter number, Y = formula number within the chapter)
- Numbering restarts within each main chapter
- Variables used in a formula must be explained directly below the formula in tabular form

0.1.9 Appendix

- Technical drawings, extensive tables, supplementary documents, etc. go into appendices
- Each appendix starts on a new page and may span multiple pages
- Appendices are numbered with consecutive uppercase Latin letters (Appendix A, Appendix B, etc.)

0.1.10 Eidesstattliche Erklärung (Statutory Declaration)

- A signed declaration of independent authorship must be included on a separate page without a page number, placed after the cover page
- It must be signed by hand

0.2 Thesis Structure and Content

0.2.1 Required Document Order

1. Deckblatt (Cover Page)
2. Erklärung des Kandidaten (Statutory Declaration)
3. Inhaltsverzeichnis (Table of Contents)
4. Abbildungsverzeichnis (List of Figures)
5. Tabellenverzeichnis (List of Tables)
6. Abkürzungsverzeichnis (List of Abbreviations)
7. Einleitung und Zielstellung (Introduction and Objective)
8. Grundlagen / Stand der Technik (Background / State of the Art)

9. Handlungsbedarf (Research Gap / Need for Action)
10. Methodenteil (Methodology)
11. Ergebnisteil (Results)
12. Diskussionsteil (Discussion)
13. Zusammenfassung und Ausblick (Summary and Outlook)
14. Literaturverzeichnis (Bibliography)
15. Anhang (Appendix)
16. Lebenslauf (Curriculum Vitae)

0.2.2 Page Count Guidelines

- Bachelorarbeit: ca. 60–80 pages
- Projekt-/Studienarbeit: ca. 60–80 pages
- Master-/Diplomarbeit: ca. 80–120 pages
- These counts exclude appendices and directories

0.2.3 Red Thread (Roter Faden)

- A clear red thread must run through the entire thesis, starting from the central research question in the introduction
- The reader must always be able to see how any part of the thesis relates to the research question
- The thesis must proceed purposefully and not lose itself in tangential topics
- All claims and statements must be causally justified; results must not merely be listed but convincingly explained
- Each main chapter must begin with a short introduction motivating the content and end with a transition to the next
- Verify the red thread by checking all headings alone (without reading the text) — the logical flow should be recognizable

0.2.4 Chapter: Einleitung und Zielstellung (Introduction)

- Length: approximately 2 pages, maximum 3 pages

- Purpose: establish contact with the reader; provide orientation ("What can the reader expect?")
- Content requirements:
  - Introduce the topic and its general background; explain why it is relevant
  - Present the specific research question
  - Justify the scope and boundaries of the thesis (what is included, what is excluded, and why)
  - Optionally describe the state of research (to show what exists and where gaps are)
  - Explain the methodological approach
  - Outline the structure of the thesis for the reader

0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)

- Clarify central concepts and define relevant terms
- Present the current state of the art objectively and without judgment (evaluation happens in the next chapter)
- Only cover theories, definitions, and knowledge directly relevant to the research question; avoid overly broad coverage
- If multiple competing theories exist, provide an overview
- Definitions are only needed for terms not commonly known by domain experts
- Present content in your own words; near-verbatim copying or translation from sources counts as insufficient
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers over websites
- Avoid popular-science sources (e.g., Schülerduden, Brockhaus); only quote newspapers for current events
- Internet sources are acceptable only if no current literature exists elsewhere (e.g., software documentation); verify their credibility

0.2.6 Chapter: Handlungsbedarf (Research Gap)

- Critically discuss the literature from the previous chapter regarding unsolved problems and errors
- Derive current deficits and research gaps; formulate the specific research question
- Explain who considers the problem significant and what its broader impact is

- Describe the contribution the thesis makes to both research (theory, models, methods, facts) and practice
- Narrow the problem statement into a concrete, achievable objective with clear expected outcomes

0.2.7 Chapter: Methodenteil (Methodology)

- Together with Results and Discussion, this forms the main body (2/3 of the thesis)
- Describe the investigation plan, procedures, and all steps and tools in sufficient detail for reproducibility
- Justify why specific methods were chosen over alternatives (pros/cons analysis)
- For experimental studies: describe the population and sample in all relevant characteristics; justify sample adequacy
- Scientific criteria for methods: objectivity, reliability, and validity
- The description must enable other researchers to reproduce the investigation
- Only describe analysis procedures in detail if they are novel or non-standard

0.2.8 Chapter: Ergebnisteil (Results)

- Present results of the investigation objectively and without interpretation
- Structure the chapter logically, aligned with the theory and methodology sections
- Use figures, tables, and diagrams to visualize results
- No interpretation or evaluation at this stage

0.2.9 Chapter: Diskussionsteil (Discussion)

- Summarize and critically discuss the results
- Evaluate and interpret results in relation to the original problem and objective from the introduction
- Argue the significance for scientific discourse and practical implications

0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)

- Maximum 3 pages
- Reconnect to the central problem and objective from the introduction
- Compactly summarize the most important results

- Place results in a broader context
- Address:
  - What are the key findings with respect to the research questions?
  - How can the results be interpreted or evaluated?
  - What questions remain open, and why?
  - What do the results mean for future work on this topic?
  - Where are opportunities for further research?
- Discuss the limitations of the investigation
- Provide suggestions for future work and improvements based on newly emerged questions

0.3 Scientific Writing Style

0.3.1 General Principles

- Write in present tense (Präsens) and nominal style (Nominalstil)
- Use impersonal form throughout; never use "ich" (I) or "man" (one/you)
- Develop the red thread of argumentation before formulating sentences
- Every sentence and every claim must be precise and well-placed
- Write result-oriented: "Die Einflüsse führen zu Veränderungen." not "Im Rahmen der Arbeit wurden untersucht…"

0.3.2 Sentence Construction

- Use passive voice sparingly — prefer active formulations
- Avoid sentences with "man" entirely
- Be careful with infinitive constructions
- Avoid chaining too many nouns together ("Substantivaneinanderreihungsproblem")
- No nested or overly complex sentences (Schachtelsätze)

0.3.3 Grammar

- Avoid relative time references ("kürzlich", "später", "modern") since the thesis should remain valid for years
- Do not use subjunctive mood (könnte, sollte, müsste, dürfte)

0.3.4 Technical Language

- Use precise adjectives; avoid vague terms like "innovativ" or "intelligent"
- No pleonasms (redundant word pairs)
- Do not artificially inflate terms
- Be cautious with superlatives and comparative forms
- Do not invent words
- Use anglicisms and foreign words sparingly; only when no equivalent German term exists
- Use precise technical terminology; avoid colloquial language
- Use technical terms consistently throughout (e.g., always "CAD-Modell", never alternating with "CAD Modell")

0.3.5 Expression and Phrasing

- Spell out numbers from zero to twelve; use digits from 13 onward
- Avoid colloquialisms and journalistic clichés
- Do not use rhetorical questions; make clear statements instead
- Avoid vague qualifiers: "im Allgemeinen", "in der Regel", "beispielsweise", "teilweise"
- Avoid filler words: "nämlich", "also", "so", "eben"
- Avoid repetition of both words and arguments
- Always formulate concretely and precisely; define unambiguously; enumerate completely; avoid generalizations
- Use nominal style: instead of "wird bearbeitet", write "Zur bedarfsgerechten Bearbeitung ist auf die Einhaltung von xy zu achten."

0.3.6 Content Guidelines

- Do not document iterative development processes or suboptimal intermediate solutions; present only the final approach
- Do not describe device/hardware development in excessive detail; focus on scientifically relevant, generalizable, transferable aspects
- Use direct quotes rarely and only pointedly; prefer evaluating and developing ideas further in your own words
- Introduce abbreviations only when necessary for space; spell out on first use and include in the abbreviation index

- Emphasize scientific quality rather than engineering challenges
- Emphasize fundamental, generalizable, and transferable findings rather than application-specific details
- Instead of giving examples, enumerate completely, structure, and evaluate
- Make all enumerations and representations as complete and non-overlapping as possible (MECE principle)
- Always support claims with your own analyses or cited sources
- No implied personal opinion (e.g., “jedoch”, “überaus”) and no subjective assessments except in the Outlook section
- Where possible, summarize each chapter’s key message in a figure for quick reference

0.3.7 Introductions and Transitions Between Sections

- Every main chapter or main section must begin with a brief introduction and end with an outlook/transition to the next section
- Example introduction: “Hasen und Igel sind, betrachtet man ihre Laufgeschwindigkeit, auf einem vollkommen unterschiedlichen Niveau.”
- Example transition: “Deshalb wird im folgenden Abschnitt der Wettlauf zwischen Hase und Igel näher betrachtet.”
- When introducing detailed explanations, name all relevant points in the introductory sentence
- Bad: “Im Folgenden werden die beiden Protagonisten näher betrachtet.”
- Good: “Im Folgenden werden die beiden Protagonisten Hase und Igel näher betrachtet.”

0.3.8 Consistency and Formatting Rules

- Introduce each abbreviation exactly once and then use it consistently
- Never write a paragraph consisting of only one sentence (minimum two sentences)
- Never write a list with only one bullet point (minimum two)
- Use nominal style for bullet-point items
- Maintain consistent spelling of expressions throughout (e.g., always “Aerosol-Jet-Druck”, never “Aerosoljet-Druck”)

0.4 Citations and Bibliography

0.4.1 Citation Style

- Citation style: DIN ISO 690 (FAPS custom style via `bibstyleFAPS.bst`)
- Reference sources in the text using numbers in square brackets: [xy]
- Page numbers are not required in citations
- Multiple consecutive citations use separate brackets: [xy] [yz], not [xy; yz]
- The bibliography is ordered chronologically and numbered consecutively; no grouping by source type
- Only actually cited works appear in the bibliography

0.4.2 Citation Placement

- A citation before the period refers to the specific sentence [xy].
- A citation after the period of the last sentence refers to the entire paragraph. [xy]

0.4.3 Direct Quotes

- Must be letter-perfect and character-perfect reproductions
- Enclose in quotation marks; cite the source at the end: "..." [20]
- Quote foreign-language texts in the original language; do not translate
- Mark own additions or rearrangements with square brackets
- Mark omissions with [...]
- Quotes longer than three lines: indent as a block, set in italic, omit quotation marks
- Do not correct errors in the original; mark them with [!] or [sic!]
- Emphasis within quotes: append [Hervorhebung durch Verfasser]

0.4.4 Indirect Quotes

- Paraphrased ideas from other sources; not placed in quotation marks
- Still require a source reference in the same placement rules as above

0.4.5 Secondary Citations

- Always prefer citing the original source

- Use secondary citations only when the original is unavailable; reference the secondary source in the bibliography

0.4.6 Plagiarism

- Provide a source reference for every piece of knowledge that goes beyond common knowledge
- Originality and independence are the most important quality criteria
- Even when rephrasing or summarizing, the source must be cited
- It must be unambiguously clear to the reader at all times what is borrowed intellectual property

0.4.7 Source Quality Requirements

- Every source must be: (1) citeable (publicly accessible), (2) citeworthy (scientific quality), and (3) relevant
- Prefer English-language sources
- Prefer journal articles and conference papers over websites
- Avoid websites as sources wherever possible; especially avoid Wikipedia, hausarbeiten.de, and similar
- If no conventional sources exist (e.g., for software topics), internet sources are permissible with critical evaluation
- Prioritize current sources from renowned journals or scientists over older work
- Quality indicators: scientific question, clear structure, precise terminology, theoretical foundation, appropriate methodology

0.4.8 Electronic Sources

- Books, journals, and publications that also exist in print are not "electronic sources" even when accessed online
- Each source entry must answer: Who? What? Where? When?
- Use DOI when available; otherwise provide the full URL and access date
- Do not insert hyphens in URLs (could be misinterpreted as part of the address)
- If author is unknown: use "o. V." (ohne Verfasser); if date is unknown: use "o. J." or "o. D."
- Schema: Nachname, Vorname (Jahr): Titel. [ggf. weitere Angaben]. DOI oder online unter: URL (Abrufdatum). No period after DOI.

0.5 Literature Research

- Recommended databases: OPAC (FAU library), Google Scholar, IEEE Digital Library, Scopus
- Strategy: create a search table with key terms, synonyms (abstract and specific) across multiple columns
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers; avoid websites
- Submit a detailed outline and relevant literature to the supervisor within 2 weeks (for Abschlussarbeiten)

0.6 Figures and Graphics — Additional Rules

- Use a self-created graphic on the first page of each chapter or major section
- If independent creation is not possible or the graphic is assembled from multiple sources, provide a source note on the first page where it appears
- Poorly formatted, inconsistent, blurry, or illegible graphics create a bad impression; avoid copies and scans
- Recreate relevant graphics, diagrams, formulas, and tables to ensure uniform formatting and meet formal requirements
- Maintain a balanced ratio of text to graphics (e.g., avoid placing two graphics directly after each other)
- Ensure consistent design across all graphics: shapes, fonts/sizes, colors

0.7 Review Checklist Before Submission

0.7.1 Content Review

- Verify all headings form a coherent red thread (recognizable without reading the text)
- Verify transitions and introductions exist at the beginning of every main chapter
- Is the central research question clearly developed in the introduction?
- Is the relevant literature reviewed with appropriate focus (not quantity, but quality of selection)?
- Can the derivation of the target concept from the analysis be followed?
- Are strengths and development potentials considered in the solution space?

- Is the proposed solution feasible and implementable?
- Does the outlook present innovative approaches for future work?
- Are all tables and figures clear and understandable?
- Is the context of existing literature properly established?
- Is there a concise summary that still covers all essential points?
- Is the overall structure clear and comprehensible?
- Is the scope sufficient yet as concise as possible?
- Were all key work items of the thesis topic fulfilled and the objectives achieved?

0.7.2 Language and Expression Review

- Check for scientific, objective tone throughout
- Check for linguistic expression quality
- Run spell-check and grammar-check
- Have at least one other person proofread the thesis

0.7.3 Visual and Formatting Review

- Check in single-page view for equal spacing between sections
- Verify balanced ratio of text and graphics (no dense clustering of figures)
- Verify consistent design of all graphics (shapes, fonts, sizes, colors)
- Eliminate all double spaces
- Check for sensible page breaks (no orphan headings, no broken table rows)

0.8 Grading Criteria

- Systematic approach and methodology
- Personal effort and creativity
- Fulfillment of the thesis assignment
- Independence during the work
- Application of domain expertise
- Discussion and critical evaluation of results
- Presentation of the initial situation
- Thoroughness, completeness, and comprehensibility of the presentation
- External form and appearance of the thesis
- Expression and style

- Spelling and grammar
- Performance is measured as work divided by time — high quality within deadlines demonstrates strong performance

0.9 Presentation Guidelines (BA/PA)

- Duration: 20 minutes ± 10% deviation
- Bring your own laptop; have a backup on a USB drive
- Submit slides to the supervisor several days before the presentation date for review
- Follow FAPS presentation guidelines
- Dress code: at minimum business casual
- Structure:
    1. Title slide
    2. Motivation slide (between title and table of contents)
    3. Table of contents slide (all relevant topics except bibliography)
    4. Content slides
    5. Summary slide
    6. Closing slide
    7. Bibliography (hidden; only shown if asked during Q&A). Cite sources on individual slides in (Author, Year) format

1 Introduction

1.1 Motivation and Context

- Industrial textile recycling requires automated sorting by garment condition (intact, minor damage, major damage, unusable)
- Current approaches rely on manual inspection or single-robot rigid-object pipelines; neither scales to deformable textiles on conveyors
- Multi-agent robotic systems combining pick, stretch, inspect, and sort offer a path to full automation
- This thesis builds on the simulation environment and validated tendon-driven robot from the preceding project thesis (Meyer, 2026)
- Central challenge: coordinating two robots under RL for deformable cloth manipulation with vision-based classification

1.2 Problem Statement and Objectives

- Thesis problem: developing a multi-agent RL system that coordinates a tendon-driven picker and a collaborative sorting arm for condition-based textile sorting in simulation
- Research objectives:
    1. Selective sorting task: retrieve T-shirts from mixed conveyor waste
    2. Deformable cloth simulation: integrate T-shirt cloth physics (PBD/VBD)
    3. Second robot arm: stretch garments for camera-based inspection
    4. Damage classification: vision-based condition assessment
    5. Multi-agent coordination: joint policies for pick-stretch-classify-sort
    6. Sim-to-real preparation: domain randomization, ROS 2 integration pathway
- Scope: simulation-based development and evaluation; physical deployment is prepared but not executed

1.3 Thesis Outline

- Brief summary of chapters 2–7 for reader orientation
- Explicitly reference the project thesis as the foundation (simulation, robot model, basic RL tasks)

2 Background

2.1 Summary of Foundation (Project Thesis)

- Brief recap of the tendon-driven robot model, simulation environment, and RL pipeline established in the project thesis (Meyer, 2026)
- Reference rather than repeat: direct the reader to the project thesis for full details
- Summarize key results: validated robot model, reach/place task performance, workspace analysis findings
- State what is taken as given for the master thesis

2.2 Cloth and Deformable Object Simulation

- Why deformable objects break rigid-body assumptions
- Common cloth modeling approaches: mass-spring, particle-based (PBD), continuum (FEM/MPM)
- Position-Based Dynamics (PBD) in PhysX: particles, constraints, Gauss-Seidel projection, XPBD compliance
- Vertex Block Descent (VBD) / Newton Thin Deformables: implicit energy minimization, vertex-level Newton updates
- Comparison: PBD vs. VBD for cloth in RL workflows (speed, stability, stiffness fidelity)
- Isaac Sim cloth integration: particle cloth API, material parameters, collision with articulations

2.3 Multi-Agent Reinforcement Learning

- Extension from single-agent MDP to multi-agent: Dec-POMDP, centralized training / decentralized execution (CTDE)
- Multi-agent PPO variants: independent PPO (IPPO), multi-agent PPO (MAPPO)
- Shared vs. independent reward structures; cooperative vs. mixed objectives
- Communication and coordination mechanisms in multi-agent RL

- Prior work: multi-robot manipulation (e.g., bi-manual grasping, handover tasks)

2.4 Vision-Based Classification for Robotic Sorting

- Visual inspection in manufacturing and recycling contexts
- CNN-based defect detection and classification (ResNet, EfficientNet architectures)
- Sim-to-real for visual classifiers: synthetic data generation, domain randomization for textures/lighting/backgrounds
- Integration of perception with robot control: observation pipelines in IsaacLab

2.5 Prior Work in Robotic Cloth Manipulation

- Robotic cloth folding, unfolding, smoothing: key benchmarks (GarmentBench, GarmentLab, Seita et al.)
- RL approaches for deformable manipulation: reward design challenges, representation learning for cloth state
- Sim-to-real for cloth tasks: domain randomization of material properties, teacher-student distillation
- Multi-robot cloth handling: bi-manual stretching, cooperative folding
- Gap: no existing work combines tendon-driven robots, multi-agent RL, and condition-based sorting

2.6 Sim-to-Real Transfer Strategies

- Domain randomization: which parameters and how aggressively
- System identification and parameter calibration
- Teacher-student policy distillation
- ROS 2 as the deployment middleware: joint state interfaces, camera topics, action servers
- Prior sim-to-real demonstrations for manipulation tasks

3 Problem Statement and Research Gap

3.1 Critical Analysis of State of the Art

- Cloth manipulation studied primarily with rigid-joint robots; tendon-driven compliance advantages unexplored for deformable objects
- Multi-agent RL applied to rigid object manipulation but not to cloth sorting pipelines
- Vision-based classification exists for quality inspection but not integrated into a multi-agent pick-stretch-sort loop
- Sim-to-real for multi-agent deformable manipulation is an open problem

3.2 Research Gap and Justification

- Key gap: no existing system combines multi-agent RL, tendon-driven manipulation, deformable cloth physics, and condition-based classification in a single pipeline
- Scientific contribution: multi-agent MDP formulation for condition-based textile sorting; evaluation of cooperative policies for pick-stretch-classify-sort
- Practical contribution: complete simulation framework for textile recycling automation; ROS 2 integration pathway

3.3 Refined Research Question

- How can a multi-agent RL system coordinate a tendon-driven picker and a collaborative sorting arm to selectively retrieve, inspect, and sort textiles by damage condition, and what strategies enable robust sim-to-real transfer?
- Expected contributions: multi-agent sorting pipeline, cloth sim integration, classification integration, sim-to-real preparation

4 Methodology

4.1 Overview of Approach

- System architecture diagram: conveyor → tensegrity picker → handoff → sorting arm → stretch → camera → classify → sort into bins
- Development stages: (1) sorting task, (2) cloth integration, (3) second arm, (4) classification, (5) multi-agent, (6) sim-to-real prep

4.2 Sorting Task (Single-Agent Extension)

- Extend cube sorting to selective retrieval: T-shirt vs. non-target objects on moving conveyor
- MDP formulation: observation (joint states, conveyor state, object labels/features), action (joint targets + gripper), reward (correct pick, correct ignore, penalty for false pick/miss)
- Training with domain randomization (belt speed, object placement, friction)

4.3 Cloth Simulation Integration

- T-shirt cloth model: particle mesh, material parameters (mass, stretch stiffness, bending stiffness, friction, damping)
- PBD vs. VBD selection and justification for training throughput
- Cloth-articulation interaction: gripper grasping cloth particles, collision handling
- Validation of cloth behavior: qualitative draping tests, quantitative stretch tests
- Domain randomization of cloth parameters

4.4 Second Robot Arm Integration

- Arm selection and kinematic configuration
- Scene placement: workspace complementarity with tensegrity picker

- Stretching primitive: grasp cloth edge, extend to flat configuration
- Camera mounting and field-of-view design

4.5 Camera-Based Damage Classification

- Simulated camera setup: resolution, positioning, synthetic image generation
- Damage classes: intact, minor damage, major damage, unusable
- Classifier architecture: CNN (e.g., ResNet-18) trained on synthetic stretched-cloth images
- Training data generation: procedural damage textures, domain randomization (lighting, background, cloth color)
- Integration as observation in the RL loop or as a post-stretch evaluation module

4.6 Multi-Agent RL Formulation

- Agent decomposition: Agent 1 (tensegrity picker), Agent 2 (sorting arm)
- Observation spaces per agent; shared state components
- Action spaces per agent
- Reward structure: cooperative (shared sorting reward), individual shaping rewards per agent
- Coordination protocol: sequential (pick → handoff → stretch → classify → sort) or learned timing
- Algorithm: MAPPO or IPPO via skrl; centralized critic with decentralized actors
- Curriculum: single-agent sub-tasks → coordinated full pipeline

4.7 Sim-to-Real Transfer Preparation

- Domain randomization strategy: systematic parameter sweep and sensitivity analysis
- ROS 2 architecture: joint state publisher/subscriber, policy inference node, camera pipeline, safety constraints
- Action-space compatibility: simulation actions → ROS 2 joint trajectory commands
- Latency and timing considerations for real-time execution
- Gap analysis: what remains to be addressed for physical deployment

5 Results

5.1 Sorting Task Results (Single Agent)

- Sorting accuracy: T-shirt correctly retrieved, non-targets correctly ignored
- False-pick rate, missed-pick rate
- Learning curves; comparison with project thesis cube sort baseline
- Effect of domain randomization on robustness

5.2 Cloth Simulation Results

- Cloth behavior validation: draping, stretching, grasping fidelity
- Performance impact: training FPS with cloth vs. rigid objects
- Stability under domain randomization of cloth parameters

5.3 Multi-Agent Sorting Pipeline Results

- End-to-end sorting accuracy by damage class
- Cycle time: pick to sort completion
- Coordination efficiency: handoff success rate, idle time
- Learning curves for multi-agent vs. single-agent baselines
- Qualitative policy behavior: keyframe sequences of the full pipeline

5.4 Damage Classification Results

- Classification accuracy per damage class (confusion matrix)
- Effect of cloth stretching quality on classification accuracy
- Robustness under visual domain randomization

5.5 Ablation Studies

- Without stretching: classification accuracy on unstretched cloth
- Without classification: sorting by pick heuristics only
- Single-agent vs. multi-agent: performance comparison
- Shared vs. independent reward: coordination quality

5.6 Sim-to-Real Readiness Assessment

- Robustness metrics under maximal domain randomization
- ROS 2 interface validation in simulation (latency, message integrity)
- Identified gaps and risk assessment for physical deployment

6 Discussion

6.1 Evaluation of Outcomes

- Achievement of each thesis objective
- Multi-agent vs. single-agent: when is coordination worth the complexity?
- Cloth simulation fidelity: sufficient for policy learning? What artifacts affect results?

6.2 Comparison with Related Work

- Comparison with GarmentBench/GarmentLab benchmarks
- Comparison with single-robot cloth manipulation approaches
- Novelty: first multi-agent tendon-driven cloth sorting pipeline

6.3 Limitations

- Cloth simulation approximations (PBD/VBD vs. real fabric behavior)
- Synthetic damage textures vs. real-world garment damage
- No physical deployment; sim-to-real gap remains
- Classification limited to synthetic training data
- Computational cost of multi-agent training with cloth

6.4 Practical Implications

- Applicability to textile recycling industry
- Modularity: individual pipeline components (classifier, sorting policy) can be updated independently
- ROS 2 integration pathway enables incremental physical deployment

7 Conclusion

7.1 Conclusion

- Summary of the complete multi-agent textile sorting pipeline
- Key quantitative achievements: sorting accuracy, classification accuracy, cycle time
- Answer the research question

7.2 Outlook

- Physical deployment via ROS 2 on real hardware
- Real-world fine-tuning: sim-to-real domain adaptation, real cloth training data
- More garment types and damage categories
- Higher-level task planning for continuous conveyor operation
- Integration with upstream waste detection and downstream recycling processes
- More realistic tendon model incorporating cable dynamics